DE LA SALLE UNIVERSITY-DASMARINAS
COLLEGE OF INFORMATION AND COMPUTER STUDIES
INFORMATION TECHNOLOGY DEPARTMENT

COURSE SYLLABUS

COURSE CODE	:	S-ITCS123
COURSE TITLE	:	Intermediate Programming
COURSE TYPE	:	Lecture
COURSE CREDIT	:	Two units
LEARNING MODALITY	:	Traditional Blended Learning
PRE-REQUISITES	:	S-ITCS112/S-ITCS112LA
CO-REQUISITES	:	S-ITCS123LA Intermediate Programming Lab
PROFESSOR	:	ROCHELLE D. SERMAÑA
CONSULTATION HOURS	:	MONDAY : 1300-1500 (ONLINE) TUESDAY: 0700-0900 (COS100-C) THURSDAY : 1300-1400 (COS100-C) 1400-1600 (ONLINE)
CONTACT INFORMATION :		rdsermana@dlsud.edu.ph (02) 7795180 / (046) 4811900 local 3134

VISION

A leading Catholic university that inspires excellence and drives innovation toward a just, peaceful, and sustainable society.

MISSION

To champion the Human and Christian Education of lifelong learners who value history and culture through responsive and inclusive academic, research, and extension programs.

COURSE DESCRIPTION:

Includes review of control structures and data types with emphasis on structured data types. Application of object-oriented programming paradigm, focusing on the definition and use of classes along with the fundamentals of object-oriented design is also part of the discussion as well as basic analysis of algorithms, searching and sorting techniques.

COURSE LEARNING OUTCOMES:

By the end of the semester, the students can:

CLO1. understand programming concepts;
CLO2. implement file processing;
CLO3. examine program functionality;
CLO4. develop cleaner, more efficient and more manageable programs in solving problems;
CLO5. develop graphical user interfaces (GUI) for Java programs;
CLO6. evaluate searching and sorting algorithms;

CLO7. apply Java's object-oriented programming concepts in the writing and editing of various levels of applications; and
CLO8. demonstrate 21st century skills in all learning activities.

LEARNING PLAN:

Learning Outcomes for Midterm Period

CLO1	Understand programming concepts	TLO1	Review basic control structures
		TLO2	Understand syntax and semantics of new programming constructs
		TLO3	Differentiate structured and unstructured data
		TLO4	Discuss how data is stored and its use
		TLO5	Explain operations that can be performed on a given structure
		TLO6	Observe how file processing works.
CLO2	Implement file processing	TLO7	Read data from a file
		TLO8	Write data to a file
		TLO9	Perform operations on a file
CLO3	Examine program functionality	TLO10	Evaluate program execution by tracing program output.
		TLO11	Test correctness of programs against several test data
		TLO12	Validate input of the program
CLO4	Develop cleaner, more efficient and more manageable programs in solving problems	TLO13	Choose an appropriate data and programming structure
		TLO14	Construct a program to implement specific searching and sorting algorithm
		TLO15	Construct user-defined methods
		TLO16	Construct recursive methods
CLO5	Develop graphical user interfaces (GUI) for Java programs.	TLO17	Design GUIs
		TLO18	Configure GUI properties
		TLO19	Perform event-driven programming
CLO8	Demonstrate 21st century skills in all learning activities.	TLO29	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem-Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	Week	Teaching-Learning Activities / Assessment Strategy	Technology Enabler	Onsite / F2F	Asynchronous	Allotted Hours
Module 0	Week 1 Jan 20-25	• Gospel Reading, Reflection, and Prayer • Course Introduction • Discussion of Syllabus • Class Participation 1 : Program Tracing on Java Fundamentals	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2
Module 1 CLO1 CLO3 CLO4 CLO5 TLO1 TLO2 TLO13 TLO17 TLO19	Week 2 Jan 27-Feb 1	• Gospel Reading, Reflection, and Prayer • Discussion of Module 1 : Review of Control Structures, Data Types and Introduction to Structured Data Types • Class Participation 2: Seatwork on Module 1	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2
	Week 3 Feb 3 - 8	• Gospel Reading, Reflection, and Prayer • Enabling Assessment 1 Online Quiz on Module 1	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2

Module 2 CLO1 CLO2 TLO2 TLO6 TLO10 TLO13	Week 4 Feb 10-15	• Gospel Reading, Reflection, and Prayer • Discussion of Module 2: File Processing • Enabling Assessment 2 : Quiz on Module 2	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2
Module 3 CLO1 CLO3 CLO5 TLO2 TLO13 TLO15 TLO17 TLO19	Week 5 Feb 17-22	• Gospel Reading, Reflection, and Prayer • Discussion of the Module 3: User-Defined Methods and Recursive Methods (User-Defined Methods Lesson Part) • Class Participation 3 : Seatwork on User-Defined Methods	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2
	Week 6 Feb24-Mar 1	Gospel Reading, Reflection, and Prayer • Enabling Assessment 3 : Short Quiz on Module 3 User-Defined Methods	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2
	Week 7 Mar 3-8	• Gospel Reading, Reflection, and Prayer • Discussion of Module 3 : Recursive Methods Part • Class Participation 4: Activity on Recursive Methods	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2

	Week 8 Mar 10-15	• Gospel Reading, Reflection, and Prayer • Enabling Assessment 4: Quiz on Recursive Methods	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2
Module 1 to 3	Week 9 Mar 17-22	***Summative Assessment:*** ***MIDTERM EXAM*** Quiz Type, Application Type covering Module 1, 2, 3		✓		2
TOTAL				***10***	***8***	***18***

Learning Outcomes for Final Period

CLO1	Understand programming concepts	TLO1	Review basic control structures
		TLO2	Understand syntax and semantics of new programming constructs
		TLO3	Differentiate structured and unstructured data
		TLO4	Discuss how data is stored and its use
		TLO5	Explain operations that can be performed on a given structure
		TLO6	Observe how file processing works.
CLO3	Examine program functionality	TLO10	Evaluate program execution by tracing program output.
		TLO11	Test correctness of programs against several test data
		TLO12	Validate input of the program
CLO4	Develop cleaner, more efficient and more manageable programs in solving problems	TLO13	Choose an appropriate data and programming structure
		TLO14	Construct a program to implement specific searching and sorting algorithm
		TLO15	Construct user-defined methods
		TLO16	Construct recursive methods
CLO5	Develop graphical user interfaces (GUI) for Java programs.	TLO17	Design GUIs
		TLO18	Configure GUI properties
		TLO19	Perform event-driven programming
CLO6	Evaluate searching and sorting algorithms	TLO20	Explain how searching and sorting algorithm works

		TLO21	Compare and contrast the different searching and sorting algorithms
		TLO22	Decide which algorithm to use in a specific problem
CLO7	Apply Java's object oriented programming concepts in the writing and editing of various levels of applications.	TLO23	Create user-defined classes
		TLO24	Create static and non-static data and methods
		TLO25	Apply access modifiers
		TLO26	Implement UML Diagrams in designing user-defined classes
		TLO27	Create array of classes
CLO8	Demonstrate 21st century skills in all learning activities.	TLO29	*Collaboration, Skilled Communication, Knowledge Construction, Self-Regulation, Real- World Problem- Solving, Use of ICT (not part of topic, but integrated in assessments)*

Module No and Learning Outcomes	**Week**	**Teaching-Learning Activities / Assessment** Strategies	**Technology Enabler**	**Onsite / F2F**	**Asynchronous**	**Allotted Hours**
Module 4 CLO1 CLO3 CLO5 CLO7 TLO2 TLO10 TLO13 TLO15 TLO17 TLO19 TLO23 TLO27	Week 10 Mar 24-29	• Gospel Reading, Reflection, and Prayer • Discussion of Module 4: User Defined Classes Part 1 • Class Participation 1: Activity on User Defined Classes Part 1 **SUSTAINABILITY WEEK**	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2
	Week 11 Mar31-Apr 5	• Gospel Reading, Reflection, and Prayer	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2

		• Discussion of Module 4: User Defined Classes Part 2 • Enabling Assessment 1: Quiz on User-Defined Classes Part 2				
	Week 12 Apr 7-12	• Gospel Reading, Reflection, and Prayer • Enabling Assessment 2: Online Quiz on User-Defined Classes	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2
	Week 13 Apr 14-19	***Independent Learning Days*** ***(April 14-15) Holy Week (April 16-19)***			✓	2
Module 5 CLO1 CLO3 CLO6 TLO02 TLO10 TLO13 TLO22	Week 14 Apr 21-26	***Research Week*** Discussion on Module 5: Searching Algorithms Enabling Assessment 3 : Online Quiz on Searching Algorithms			✓	2
	Week 15 Apr 28-May 3	• Gospel Reading, Reflection, and Prayer • Class Participation 2: Presentation about Searching Algorithms	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2

Module 6 CLO1 CLO3 CLO6 TLO2	Week 16 May 5-10	• Gospel Reading, Reflection, and Prayer • Discussion on Module 6: Sorting Algorithms • Class Participation 3: Presentation about Sorting Algorithms	Schoolbook , PowerPoint, Java SDK & NetBeans IDE	✓		2
TLO10 TLO13 TLO22	Week 17 May 12-17	• Gospel Reading, Reflection, and Prayer • Enabling Assessment 4: Quiz on Sorting Algorithms	Schoolbook , PowerPoint, Java SDK & NetBeans IDE		✓	2
Module 4 to 6	Week 18 May 19-24	***Summative Assessment: FINAL EXAM*** Application Type (Output-Based) covering Module 4,5,6		✓		2
TOTAL				***10***	***8***	***18***

GRADING SYSTEM:

Midterm	%	Final	%
Enabling Assessments	50%	Enabling Assessments	50%
Summative Assessment	30%	Summative Assessment	30%
Class Participation	20%	Class Participation	20%
	100%		100%

FINAL GRADE = $\frac{\text{Midterm Grade + Final Grade}}{2}$

COURSE POLICIES AND REQUIREMENTS:

1. This course follows the Traditional Blended Learning Mode. There will be 50% onsite and 50% asynchronous.

2. Enrollment in an E-Class. The Center for Innovative Learning Programs (CILP) shall enroll you in the e-class. This will be based on the class list provided by the Office of the University Registrar and at the portal.

3. Communication and Feedback
    a. Onsite classes shall happen as indicated in the learning plan section of the syllabus. Fifty percent (50%) of the weekly classes is delivered face-to-face while the remaining is asynchronous.
    b. Online Asynchronous Communication shall happen when the faculty utilizes at least one of the recommended asynchronous communications. The minimum response time for asynchronous communication is 48 hours, excluding Sundays and holidays. All communication platforms must be explicitly indicated in the syllabus. Below are the recommended asynchronous communication platforms:
        i. Forum – at the Forum Tab at Schoolbook, students can post their general concerns. The forum is public, and any post can be seen by the class.
        ii. Messaging – at Schoolbook. Any private or confidential matters may be messaged through the Schoolbook messaging feature.
        iii. Email – you may email your professor. (Professor's email will be given during the course orientation)
    c. No use of foul, curse, and other bad words, whether written or verbal. No shouting. No flooding of chat and messages. No selling.

4. Attendance Policy
    a. Attendance Policy for Onsite Classes is scheduled based on the learning plan. A maximum of 20% for class participation/recitation/attendance shall be allotted for the student's grade.
    b. The onsite meeting will only be held at the schedule indicated in the registration form.
    c. Students are required to attend onsite classes, 40% of absence (from onsite classes) for the entire semester means 0.0 final grade.

5. Intellectual Property Policy
    a. Policy and copyright infringement will not be tolerated. Any violation of such will earn students a grade of zero in the assessment.
    b. Using any illegally obtained software is strictly prohibited.
    c. Students are encouraged to use their original photos, videos, and other resources. Otherwise, students can use royalty-free resources or embed the sources in their submissions to avoid copyright infringement and/or plagiarism.
    d. Giving passwords to Schoolbook and Office 365 is strictly prohibited. Likewise, accessing Schoolbook and Office 365 accounts other than the students' personal account is also strictly prohibited. Violating students will be reported to the Student Welfare and Formation Office (SWAFO).

6. Other Policies
    a. Attire - students must wear decent casual or smart-casual attire prescribed by the Student Handbook.
    b. Students must be honest at all times; cheating and plagiarism in any form will merit a grade of 0.00.
    c. Cellular/Mobile phones should always be in silent mode during class hours; the use of cellular phones is prohibited in class unless special permission is sought. Cellular phones cannot also be used as calculators during examinations.
    d. Any complaints against the teacher or against classmates should be properly addressed to the subject teacher for appropriate action. Students may seek the help and guidance of

their academic/registration adviser in resolving the issue with the subject teacher when such mediation is needed.

REFERENCES

Onsite References

Provider	Reference Material
AEA Lib QA 76.73.J38 .Sch35 2022	Schildt, H .2022. Java : a beginner's guide.(New York): McGraw-Hill Education

Online References

Call number or e-provider	Reference Material
eBook	Prudhomme, G. 2020. *Java Programming Applications*. Arcler Press.
Web Resource	Java Classes and Objects. c1999-2024. Refsnes Data. [accessed 2025 Jan 29]. https://www.w3schools.com/java/java_classes.asp
Web Resource	Searching Algorithms.GeeksforGeeks.Sanchhaya Education Private Limited. [accessed: 2025 Jan 29].https://www.geeksforgeeks.org/searching-algorithms/
Web Resource	Sorting Algorithms.GeeksforGeeks.Sanchhaya Education Private Limited. [accessed: 2025 Jan 29].https://www.geeksforgeeks.org/sorting-algorithms/
Website	ProgrammingKnowledge. Java Tutorial for Beginners (Step By Step Tutorial). YouTube.[accessed 2025 Jan 29]. https://m.youtube.com/playlist?list=PLS1QulWo1RIbfTjQvTdj8Y6yyq4R7g-Al

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